

Project 01: Data Cleaning & Preprocessing

Step 1: Strategic Imputation (Resolving Missing Values)

Objective

The **CouponCode** column initially contained **309 missing values**. To preserve all **1,200 records**, maintain **data integrity**, and ensure dataset consistency, **Mode Imputation** was applied by replacing missing values with the most frequently occurring coupon code (**Freeship**).

Frequency Analysis (Before Imputation)

Coupon Code	Count
Freeship	313
Save10	286
Winter15	292
Blank (Missing)	309
Total Records	1,200

Technical Execution

Excel Formula

`=IF(ISBLANK(A2),"Freeship",A2)`

Finalization: Converted formulas into permanent values using **Paste Special → Values**.

Modification Log

Description	Impact	Status
Imputed 309 missing values using Mode Imputation (Freeship)	Preserved 100% of records (1,200/1,200)	✓ Completed

Result

- ✓ 309 missing values successfully resolved.
- ✓ Freeship count increased from **313** to **622**.
- ✓ 100% of records preserved.

Step 2: Integrity Audit (Duplicate Verification)

Objective

To verify that each record represents a unique transaction, the **Order ID** column was audited for duplicate values to ensure data accuracy and prevent duplicate transaction counts.

Technical Execution

- Applied **Conditional Formatting** to highlight duplicate **Order ID** values.
- Used **Data** → **Remove Duplicates** with **Order ID** selected as the primary key.

Audit Result

Verification	Result
Total Records	1,200
Duplicate Order IDs	0
Unique Order IDs	1,200

Modification Log

Description	Impact	Status
Verified duplicate values in the Order ID column	Confirmed 100% unique transactions	 Completed

Result

- No duplicate Order IDs found.
- Verified 1,200 unique transaction records.
- Dataset integrity successfully maintained.

Step 3: Date Standardization & Chronological Sorting

Objective

To standardize the **Order Date** format and organize all records chronologically, ensuring consistency and improving data quality for analysis.

Technical Execution

Date Format Conversion

Excel Formula

`=TEXT (A2, "yyyy-mm-dd")`

Sorting: Applied **Ascending Order (Oldest to Newest)** to arrange all **1,200 records** chronologically from **2023 to 2025**.

Modification Log

Action Taken	Technique	Impact
Standardized date format	<code>=TEXT (cell, "yyyy-mm-dd")</code>	Created a consistent YYYY-MM-DD format
Sorted records	Ascending Order	Organized records chronologically (2023–2025)

Result

- All dates standardized to **YYYY-MM-DD** format.
- 1,200 records sorted chronologically.
- Dataset consistency and readability improved.

Step 4: Text Standardization (TRIM & PROPER)

Objective

To standardize text values by removing extra spaces and correcting capitalization, ensuring consistency across all categorical columns.

Technical Execution

Excel Formula

`=PROPER (TRIM (A2) )`

Applied To

- Product

- City
- Status
- Customer ID

Modification Log

Change ID	Description	Impact	Status
CR004	Applied =PROPER (TRIM ()) to categorical columns	Removed extra spaces and standardized text formatting	 Completed

Result

- Extra spaces removed from text values.
- Capitalization standardized across categorical columns.
- Consistent and clean text formatting achieved.

Step 5: Numeric Standardization (ROUND Function)

Objective

To standardize numeric values by removing unnecessary decimal places, ensuring consistent and accurate financial data.

Technical Execution

Excel Formula

=ROUND (A2 , 2)

Applied To

- Unit Price
- Total Price

Modification Log

Description	Impact	Status
Applied =ROUND () to Unit Price and Total Price	Standardized decimal values and improved numeric consistency	 Completed

Result

- Unit Price and Total Price standardized to **two decimal places**.
- Improved numeric consistency and reporting accuracy.